

Lab 2: Multi-Hop Communication

General Instructions

In the previous lab, we studied Bluetooth Low Energy (BLE) single-hop communication, which enables local, short-range data exchange between IoT devices. In this lab, we will explore multi-hop communication, where data is relayed through intermediate nodes to extend the communication range and enable connectivity in larger networks. For simplicity, we limit ourselves to a line topology.

Your group can get a total of 10 points in this lab: 8 points for the implementation and evaluation, 2 points for the presentation. Remember that those groups that will not present this lab at the lab session have to upload a 6 to 8 min long video presentation with their submission in Moodle.

Please note that due to whit monday, the deadline for this lab is extended to **May 31, 2026 at 23:59**.

Part 1 – Network Formation (2 points)

First, your task is network formation. Essentially, you should form a network that you can use in the second step for collecting sensor data.

Essentially, your task is to extend your existing code to allow the multi-hop propagation of a button press. For this part, consider a line topology with 4 nodes, each of them 10 meters apart. When you press a button on an end device, this device initiates the formation of a network and is the only device where button presses have an effect until a reboot of the devices. After forming the network, pressing the button shall turn on a LED on this board and on all the other boards, and releasing the button shall turn off all LEDs.

Part 2 – Evaluation and Discussion of Part 1 (2 points)

Critically assess your implementation and reflect on the design choices and potential improvements. Some questions that you could answer in this context are:

- What Latency do you observe for your message propagation? How does it change with the number of hops?
- How reliable is your implementation? Do you observe any message losses? If so, how do they affect the overall performance of the network?
- Does your implementation scale well with the number of nodes? What challenges do you foresee when scaling to larger networks?

- Is your design limited to a line topology, or can it be adapted to other topologies (e.g., mesh)? What modifications would be necessary to support different topologies?

You are not limited to these questions, they only serve as orientation.

For Renode, please use and modify the attached simulation script `lab_2.resc`. This script exposes four UART terminals `term_1`, `term_2`, `term_3`, and `term_4` via TCP sockets¹. You can use, e.g., Python to read data from them to process it further. You can also take a look at `pyrenode3` to manage your simulation from Python and have direct access to the output for your evaluation. In the simulation script, you have to adjust the path to the location of the attached modified board description to ensure that not all BLE devices have the same device address. You shall run your evaluation both with a range-based wireless medium without loss (line 22 in `lab_2.resc`), and with loss (line 23). Comment in the respective line.

Part 3 – Measurement Collection (2 points)

In lab 1, you formed a multi-hop network. Now we want to use this network for collecting sensor data from each node to the sink node. However, instead of sending actual measurements, you will send random temperature and humidity values. The temperature values should lay in the range between -25.0 and 200.0 degrees Celsius, and the humidity values between 0.0% and 100.0% . Your random number generator shall generate values within the given intervals, including the limits of the interval with one decimal place. (*Hint: You can use fixed-point numbers*².)

The protocol you implement shall behave the following. When you press a button on an end device (sink), this device initiates the formation of a network. Any other button presses on any device shall not have any effect until a reboot of the devices. After network formation, all four devices in your network shall perform/generate a measurement every 200ms, timestamp this measurement and send the measurement, the timestamp, the node's node-ID, and the value of a local measurement counter to the sink. The sink, which also generates and timestamps measurements, collects all the measurements and prints them with the timestamp and the transmission time to the serial interface. To be able to calculate the transmission time, the devices need to be time-synchronized. Instead of implementing a time-synchronization algorithm, you can press the reset button³ on all of them simultaneously to start all of them at the same time.

Each measurement shall be printed as a single line in this format:

`<nodeID>;<measurement-counter>;<temp>;<humidity>;<timestamp>;<tx-time>`

¹<https://renode.readthedocs.io/en/latest/host-integration/uart.html>

²https://en.wikipedia.org/wiki/Fixed-point_arithmetic

³You might have to enable the reset button first: https://infocenter.nordicsemi.com/index.jsp?topic=/ug_nrf52840_dk/UG/dk/boot_reset_but.html

Part 4 – Evaluation and Discussion of Part 3 (2 points)

To evaluate the reliability and latency of your system, you will use Renode. Instead of pressing a button, you can program a specific node with a slightly modified firmware that initiates the network formation on start.

You may output intermediate outputs at different nodes, but what counts is the output at the sink node in the form given above. With the data printed at the source node, you shall analyze latency, reliability and percentage of sensor readings received in time. Please use plots for your evaluation showing reliability and latency distribution for each node in the network.

For Renode, please use and modify the attached simulation script `lab_2.resc`.

As a first step, evaluate your system with four nodes. As a second step, evaluate your system twice present in Renode (comment in lines 12, 66–88, 115–125 in `lab_2.resc`). For the second step, ensure that only the four nodes belonging to the same line associate and communicate with each other. For both steps, your evaluation shall contain 500 measurement readings at each node. Moreover, you shall run both evaluation steps with a range-based wireless medium without loss (line 22 in `lab_2.resc`), and with loss (line 23). Comment in the respective line.